

GSE32863 LUAD data audit

The [GEO GSE32863](#) Series Matrix contains 116 log2-transformed, robust spline normalized expression samples on GPL6884. The GEO overall design describes 58 pairs, but the sample titles identify only 57 complete patient IDs. GSM813507 is 3023_T with no 3023_N; GSM813518 is 3035_N with no 3035_T. Their metadata also names different subjects, so neither is treated as the other's pair. The two samples are excluded. The analysis uses 114 samples from 57 verifiable pairs.

The matrix has 48,803 probes. GPL6884 annotation yields 28,718 probes with one unambiguous gene symbol; the predetermined maximum-IQR probe rule retains 19,404 genes. A paired tumor-minus-normal t-test with Benjamini-Hochberg adjustment and $FDR < 0.05$ plus absolute $\log_2FC \geq 1$ finds 512 upregulated and 749 downregulated genes. These are disease-signature results, not drug efficacy findings. A paired [limma](#) sensitivity run (R 4.6.1, limma 3.68.5; patient fixed effects, `lmFit + eBayes`) produced the same 512 up and 749 down thresholded genes, with Jaccard overlap 1.0 in each direction. $\log_2FC$ values agree to numerical precision, while p-values and FDRs differ because limma moderates the variance. Exact outputs and versions are in [data/manifests/gse32863-limma.json](#).

Run `python scripts/process_gse32863.py` after placing the official GSE32863 Series Matrix and GPL6884 annotation under `data/raw/GSE32863/`. For the sensitivity run, install R and Bioconductor limma, then run `python scripts/run_limma_sensitivity.py --rscrip <path-to-Rscript>`. File URLs and SHA-256 hashes, QC counts, and generated-output hashes are in [data/manifests/gse32863.json](#). Full expression matrices and derived per-gene files remain outside Git.

The GSE92742 metadata was frozen for the A549 chemical-treatment cohort: 24 hours, 10 μ M, 5,814 signatures and 5,169 unique `pert_id` values. Eleven reference drug names were saved in `configs/luad_positive_controls.csv` before any LUAD drug ranking. Five names match this exact metadata subset directly; unmatched names require ID-level review, not guessed synonym matching.

The [ExperimentHub EH3226 lincs.h5](#) is a public GSE92742-derived 10 μ M, 24-hour Level 5 subset. Its upstream `sig_filter` retains the first technical duplicate for each drug name and cell line. We verified the 2,464,124,175-byte file (SHA-256 8087bd029d29df17ce44eb653610ad0e2b847e3f361d5540e5f798da42a473ec) and selected 4,920 A549 drug-name signatures. The LUAD signature intersects 961 of the 978 LINCS landmark genes; 57 are in the prespecified up set and 50 in the down set. Negative Spearman and gene-set connectivity were fused with fixed RRF ($k=60$). See [the Top-10 screening report](#).

EH3226 stores names in its columns but no per-column `pert_id`. GEO metadata maps 206 of the 4,920 names to multiple IDs. The Broad Repurposing Hub sample audit finds only two exact InChIKey matches among the Top-10 inferred GEO IDs. Consequently, Hub targets and clinical-phase annotations remain name-level context, never chemical-identity proof. The full GEO Level 5 archive was not needed for this bounded analysis.